Xuanfei Ren

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Education

University of Wisconsin-Madison — **Madison, USA**
PhD student, advised by Tengyang Xie, Department of Computer Sciences. — *Aug. 2024 – Present*

University of Science and Technology of China — **Hefei, China**
Bachelor of Science in Mathematics, specialization in Probability and Statistics. — *Aug. 2020 – Jun. 2024*

Experience

Amazon Web Services, Annapurna Labs — **Cupertino, USA**
Applied Scientist Intern, Neuron Science team, working with Shengzhe Xu and Shaowei Zhu. — *May 2026 – Present*

- Working on **agentic kernel optimization** for AWS **Trainium** AI accelerators.

Research Interests

My research interests span **AI agents** and **reinforcement learning**, encompassing both theoretical foundations and experimental work.

Publication

- **When Does Trajectory-Level Supervision Permit Efficient Offline Reinforcement Learning?**
  Xuanfei Ren, Tengyang Xie. *Submitted to ICLR 2027.*
- **ARTISAN: Unlocking LLM Kernel Optimization on Unfamiliar Hardware**
  Yufan Du, Xuanfei Ren, Shaojie Xiang, Jason Cong. *Submitted to ICLR 2027.*
- **POLCA: Stochastic Generative Optimization with LLM**
  Xuanfei Ren, Allen Nie, Tengyang Xie, Ching-An Cheng. ***ICML 2026.***
- **Optimal Batched Linear Bandits**
  Xuanfei Ren, Tianyuan Jin, Pan Xu. ***ICML 2024.***

Research Projects

Outcome-Based Offline Reinforcement Learning

Work with Prof. Tengyang Xie

- Developed a statistical theory for **offline RL from trajectory-level supervision**: proposed the **OPAC** algorithm, which jointly learns a latent per-step reward model and optimizes policies via pessimistic actor-critic updates; proved a sharp guarantee of order $\widetilde{O}(H^2\sqrt{C_{\mathrm{sa}}(\pi^\star)/n})$ with a matching lower bound; extended the framework to **trajectory-level preference feedback**.
- Introduced a **generalized outcome-based RL** formulation, proving an exponential lower bound for all-success objectives and identifying tractable regimes via the structural coefficients.

Stochastic Generative Optimization with LLM

Work with Dr. Ching-An Cheng (Google Research), Dr. Allen Nie (Google DeepMind), and Prof. Tengyang Xie

- Formalized complex system tuning as a **stochastic generative optimization** problem, utilizing LLMs as optimizers to automate iterative refinement through numerical rewards and natural language feedback.
- Engineered a framework combining a **priority queue-based search strategy** to balance exploration and exploitation, an **ε-Net mechanism** to ensure parameter diversity, and an **LLM Summarizer** for meta-learning, achieving **provable scalability** over naive baselines under noisy evaluations.
- Demonstrated state-of-the-art performance across diverse benchmarks: **HotpotQA** (prompt tuning), **τ-bench** (agent orchestration), **VeriBench** (code translation/Lean4), and **KernelBench** (high-performance CUDA kernels).
- Contributed to OpenTrace, an open-source library for end-to-end generative workflow optimization.

Optimal Batched Linear Bandits — Duke University

Advisor: Prof. Pan Xu (Department of Computer Science, Duke University) — *Aug. 2023 – Jan. 2024*

- A theoretical work on **batched linear bandits**: devised the first algorithm to simultaneously achieve **asymptotic and non-asymptotic regret optimality** with the **least possible batch complexity**, a goal previously unattained.
- Confirmed the algorithm's superiority over existing baselines through rigorous experimentation.

Community Service

Conference Reviewer: NeurIPS, ICML, ICLR, RLC, ACL.

Skills

Programming: Python, PyTorch, CUDA, NKI (Trainium kernels), Lean 4, R, LaTeX, MATLAB, Mathematica.